

Tutorial series 1

Exercise 1:

- 1) Prove that if $r \in \mathbb{Q}$ and $x \notin \mathbb{Q}$, then $r + x \notin \mathbb{Q}$. Now, if $r \neq 0$ then $r \cdot x \notin \mathbb{Q}$.
- 2) Show that $\sqrt{2} \notin \mathbb{Q}$.
- 3) Deduce that: between any two rational numbers, there is always an irrational number.

Exercise 2: Prove by induction that:

- 1) $(1 + x)^n > 1 + nx, \quad \forall x > -1, x \neq 0, n \geq 2$
- 2) $\sum_{k=0}^n k^2 = \frac{n(n+1)(2n+1)}{6}$
- 3) $\sum_{k=0}^n k! = (n+1)! - 1$

Exercise 3: Solve in $\mathbb{R}$:

- 1) $|x - 3| = |x + 1|$
- 2) $|2x - 4| \leq |x + 2|$
- 3) $|x + 1| + |x - 3| \leq 6$
- 4) $|x + 12| \leq |x^2 - 8|$
- 5) $(x + 1)(x - 1) > (x + 1)^2$
- 6) $\lambda x + 7 \geq 3x - 5\lambda, \quad \lambda \in \mathbb{R}$

Exercise 4: Let a, b, c three real numbers, show the following inequality:

- 1) $ab \leq \frac{a^2 + b^2}{2}$
- 2) $ab + bc + ac \leq a^2 + b^2 + c^2$
- 3) $3ab + 3bc + 3ac \leq (a + b + c)^2$

Exercise 5: Let x, y two real numbers, prove that:

- 1) $\max(x, y) = \frac{1}{2}(x + y + |x - y|)$
- 2) $\max(x, y) = \frac{1}{2}(x + y - |x - y|)$

Exercise 6: Let x, y, u, v real numbers, prove that:

- 1) $||x| - |y|| \leq |x - y|$
- 2) $||x - u| - |y - v|| \leq |x - y| + |u - v|$
- 3) $|ux - vx| \leq \sqrt{(x^2 + y^2)(u^2 + v^2)}$
- 4) $\frac{|x+y|}{1+|x+y|} \leq \frac{|x|}{1+|x|} + \frac{|y|}{1+|y|}$

Exercise 7: Let $x \in \mathbb{R}$,

- 1) Determine $E(2x)$ in function of $E(x)$.
- 2) Simplify $E(x + k)$, with $k \in \mathbb{Z}$.
- 3) Solve the equation $E(2x + 1) = E(x + 4)$.

Exercise 8: $\forall x \in \mathbb{R}, \forall n \in \mathbb{N}^*$, show that:

- 1) $0 \leq E(nx) - nE(x) \leq n - 1$
- 2) $E\left(\frac{E(nx)}{n}\right) = E(x)$

Exercise 9:

Are the following sets bounded above? bounded below? If so, determine their lower bound and upper bound.

$$A = \left\{\frac{1}{x}, x \in [1, 5]\right\}, \quad B = \left\{1 + \frac{1}{n}, n \in \mathbb{N}^*\right\}, \quad C = \left\{\frac{1}{n} + \frac{1}{p}, n, p \in \mathbb{N}^*\right\}$$
$$D = \{\sqrt{x+1} - \sqrt{x}, x \in \mathbb{Q}^+\}, \quad E = \{x^2 < 2, x \in \mathbb{R}\}$$

Exercise 10:

Let the subset A a non-empty and bounded in $\mathbb{R}$. We define the subset B by: $B = \{x - y, (x, y) \in A^2\}$.

- 1) Justify that A has an upper bound and a lower bound.
- 2) Show that $\text{Sup}(A) - \text{Inf}(A)$ is an upper bound for subset B .
- 3) Deduce that subset B has an upper bound, and then prove that $\text{Sup}(B) = \text{Sup}(A) - \text{Inf}(A)$.

Exercise 11:

Let A and B two subsets of $\mathbb{R}_+^*$. Suppose that A is bounded above, we define $E = \left\{z = \frac{x}{1+y}, x \in A, y \in B\right\}$.

- 1) Justify the fact that $\text{Sup}(A)$ and $\text{Inf}(B)$ exist.
- 2) Prove that: $\forall z \in E, z \leq \frac{\text{Sup}(A)}{1+\text{Inf}(B)}$.
- 3) Deduce that: $E = \frac{\text{Sup}(A)}{1+\text{Inf}(B)}$.